

MicroPulse — MLI & Metabolic Load Score Guide

The two pillars under Composite Load — what they measure and how to read them

MicroPulse Performance Intelligence Platform

April 2026

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1. What this is

The Composite Load score that you see on the Daily Briefing card (“Comp 0.78”) is not a single measurement — it is a blend of two underlying load engines. This guide explains both of them.

The two engines answer different questions:

- **Mechanical Load Index (MLI)** — *“How hard did this player’s joints, tendons and connective tissue work today?”* Built from accelerations, decelerations, change-of-direction events, player-load-per-minute and impacts.
- **Metabolic Load Score (MLS)** — *“How hard did this player’s energy systems work today?”* Built from metabolic power, high metabolic load distance and time spent above the high-power threshold.

Most coaches do not need to read raw MLI or MLS values day-to-day — Composite Load summarises them. But knowing what’s underneath helps you act intelligently when something goes wrong, and helps you understand the “fatigue type” tag that appears next to flagged players (“mechanical fatigue”, “metabolic fatigue”, “global fatigue”).

Quick mental model

- **MLI = mechanical wear-and-tear.** High MLI is what you feel in your knees, hips and Achilles the next day. Driven by hard braking and direction changes.
 - **MLS = energy-system fatigue.** High MLS is what you feel as breathlessness and aerobic depletion. Driven by sustained high-intensity running.
 - When **both** are high it’s a **global** day — the player is tired everywhere. Often post-match.
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2. Mechanical Load Index (MLI)

2.1 What it measures

MLI is a single 0-100 number representing how mechanically demanding today's session was for this player, **relative to the player's own 28-day baseline**. A 50 means “exactly your average mechanical load”; a 70+ means “much harder than usual”; a 30 means “much lighter”.

The MicroPulse implementation is based on the published mechanical-load monitoring literature (notably Vanrenterghem 2017 and the McBurnie 2022 deceleration framework) and combines four sub-scores into a single weighted total.

2.2 The four sub-scores under MLI

Sub-score	Stands for	Inputs	Weight in MLI
DSS	Deceleration Sub-Score	High decels, total decels, IMA decel count	40%
ASS	Acceleration Sub-Score	High accels, total accels, IMA accel count	20%
CSS	Change-of-Direction Sub-Score	IMA CoD events, CoD event count	25%
GDS	General Dynamic Sub-Score	Player-load-per-minute, IMA total, impacts	15%

The decel sub-score is weighted most heavily because eccentric braking is the highest tissue-damage driver in field sport — the McBurnie 2022 finding we built the whole Decel Intelligence engine around. Acceleration and CoD round out the picture; player-load-per-minute and impacts are the catch-all dynamic burden signal.

Each sub-score is itself a personally normalized 0-100 number (today's value divided by the player's 28-day median, capped at 2x and rescaled). So a DSS of 80 means “today's decel volume was 1.6x your normal”.

2.3 The MLI bands — what the number means

MLI score	Band	Color	What it means
0–39	LOW	Grey	Light session — well below this player's normal mechanical load
40–54	MODERATE	Green	Within personal range — typical training day
55–69	HIGH	Yellow	Hard session — meaningfully above this player's normal

70–84	VERY_HIGH	Orange	Very demanding — comparable to or exceeding match load
85+	EXTREME	Red	Extreme spike — usually only seen on tournament days or freak sessions

Why the bands are personal, not absolute

A centre-back's "normal" mechanical day might involve 12 high-intensity decels and 4 sprints. A wide forward's normal might be 22 sprints and 8 decels. If we used absolute thresholds, central players would always look "low MLI" and wide players would always look "high". By comparing each player against their own 28-day baseline, the same MLI band means the same coach action regardless of position.

2.4 Residual MLI — the 3-day rolling read

A single MLI number tells you about today. But injuries don't usually come from one hard day — they come from accumulated load that the body hasn't recovered from. Residual MLI is a 3-day rolling weighted accumulation:

Residual MLI = today $\times$ 1.0 + yesterday $\times$ 0.6 + 2-days-ago $\times$ 0.3

So a player whose MLI was 65 today, 60 yesterday and 70 two days ago has a Residual MLI of $65 + 36 + 21 = 122$.

The residual bands flag accumulated risk:

Residual MLI	Band	What it means
<70	NORMAL	Steady accumulation, no concern
70–109	ELEVATED	Multiple consecutive demanding days — monitor
110–134	CAUTION	Sustained high mechanical load — recovery time recommended
≥ 135	HIGH	Multi-day mechanical overload — modify or rest before next hard session

Residual MLI is what bumps Composite Load Concern into "moderate" or "high" even on a day where today's MLI looks normal. The body is still carrying the accumulated effect of the previous two days.

2.5 The flags that fire from MLI

Beyond the band, the MLI engine raises specific flags when individual signals spike:

- **Decel spike** — high-decel ratio $\geq 1.35 \times$ personal baseline. Most predictive single flag for hamstring strain.
- **CoD spike** — IMA CoD ratio $\geq 1.40 \times$ personal baseline. Predictive for groin and ACL.

- **Dense mechanical** — player-load-per-minute AND IMA total both $\geq 1.25\text{--}1.3\times$ baseline. Means the session was mechanically intense without being long.
- **High mechanical** — $\text{MLI} \geq 70$ (the VERY_HIGH band)
- **Extreme mechanical** — $\text{MLI} \geq 85$ (the EXTREME band)

These flags surface in the verdict explanation and the AI Player Summary when relevant.

2.6 Confidence

MLI requires up to 11 input signals (the various accel/decel/CoD/impact counts). When data is partial, the score still computes but with reduced confidence:

- **High confidence** — $\geq 80\%$ of inputs present AND ≥ 5 days of baseline. Trust the number.
- **Medium confidence** — 50–80% of inputs present. The MLI is directionally correct but specific banding may shift if more data arrives.
- **Low confidence** — $< 50\%$ of inputs. The score should be read as “rough estimate” only — usually appears for new players (still building a baseline) or when Catapult had a bad data day.

The confidence is shown next to the score in the load-detail UI when relevant.

3. Metabolic Load Score (MLS)

3.1 What it measures

MLS is a 0-100 score representing how energy-system-demanding today’s session was, relative to the player’s 28-day metabolic baseline. Built from Catapult’s metabolic-power family of fields (di Prampero 2005 framework).

Four signals contribute, each z-scored against the player’s 28-day metabolic distribution then weight-blended:

Input	Weight	What it captures
HML Distance (m)	40%	Distance covered while metabolic power exceeds the high threshold ($\sim 25.5\text{ W/kg}$)
Time above HML threshold (s)	30%	How long the player was working at high metabolic power
Peak metabolic power (W/kg)	20%	The single hardest moment of the session
Average metabolic power (W/kg)	10%	Mean intensity over the whole session

The 40% weight on HML distance is intentional — a player who covers a lot of ground at high metabolic power is doing the work that drives aerobic adaptation (and aerobic fatigue). Peak power is a sanity check, average power is the smoothed background.

3.2 The score formula

$\text{score} = \text{clamp}(50 + 15 \times (\text{weighted_z_composite}), 0, 100)$

So 50 is the player's personal mean, every full standard deviation above adds 15 to the score, capped at 100. A score of 80 means “today was about 2 standard deviations above your typical metabolic day”.

3.3 The MLS bands

MLS score	Band	What it means
0–34	LOW	Light metabolic day — well below normal
35–54	MODERATE	Within personal range — typical training day
55–74	HIGH	Demanding metabolic session — above this player's normal
75–100	VERY_HIGH	Very demanding — comparable to or exceeding match metabolic load

3.4 Important caveat about MLS — the Brown 2016 finding

The Catapult-derived metabolic power numbers that feed MLS are **systematically wrong** in specific ways (see Brown et al. 2016, J Sci Med Sport):

- **Walking** → GPS-MP overestimates true energy cost by ~43%
- **Steady running** → GPS-MP within 5–8% of truth
- **Intermittent (sport-specific)** → GPS-MP **under-estimates** true energy cost by ~44%

This means MLS will under-flag intermittent-heavy days and over-flag walk-heavy days. We addressed this by adding the `metabolic_power_corrected_brown` column with activity-aware multipliers ($\times 1.43$ for intermittent, $\times 0.566$ for walk, $\times 1.0$ for steady). The corrected number is what powers the Indoor Composite Load Score's metabolic weighting; MLS itself currently still uses raw values until we re-run baselines on corrected numbers.

If MLS reports a player at “moderate” but the Decision Summary's verdict says “watch” because of decel signals, **trust the verdict** — MLS is the noisier of the two engines for football specifically.

3.5 Metabolic confidence

Like MLI, MLS has confidence tiers based on input coverage:

- **High** — all 4 metabolic inputs present today, ≥ 10 baseline days
- **Medium** — 2–3 inputs, ≥ 6 baseline days. Score directionally correct.
- **Low** — < 2 inputs OR < 6 baseline days. Score not used in Composite Load.

Composite Load only includes MLS when confidence is medium or high. Below that, the metabolic arm is excluded and Composite is computed from MLI + ACWR alone.

4. How MLI and MLS combine into Fatigue Type

When both MLI and MLS are available with at least medium confidence, the engine categorises the dominant fatigue type. This is the “mechanical fatigue” / “metabolic fatigue” / “global fatigue” tag you see next to flagged players.

Combination	Tag	What it means	What to modify
MLI HIGH, MLS LOW–MOD	mechanical_fatigue	Joints/tendons taxed, energy systems fine	Drop decel-heavy drill; aerobic work OK
MLI LOW–MOD, MLS HIGH	metabolic_fatigue	Energy systems taxed, joints fine	Drop long-running drill; explosive work OK
Both HIGH	global_fatigue	Everything taxed — typical post-match	Drop overall intensity, prioritize recovery
Both LOW–MOD	normal	Healthy training day	Train as planned

Coaches who can quickly read the fatigue type save the player from the wrong modification. A common mistake is to “give a recovery jog” to a player with mechanical fatigue — the mechanical load doesn’t recover from a jog, it recovers from time off the legs. Conversely a metabolic-fatigued player can still benefit from short, sharp explosive work because their joints aren’t compromised.

5. Where you see MLI and MLS in the UI

Most coaches will not look up raw MLI/MLS numbers. The values feed into:

- **Composite Load Concern** (the “Comp 0.78” chip on Daily Briefing) — both engines feed in
- **Fatigue type tag** (“mechanical fatigue” / “metabolic fatigue” / “global fatigue”) on player rows
- **AI Player Summary** — synthesizes both engines into plain-language coach speak
- **Decision Summary verdict explanation** — the “Why” block lists MLI/MLS reasons when they drove the verdict

For coaches who want to drill into the raw numbers per player per day, the dev-coach-dashboard exposes them — but day-to-day, trusting the synthesised composite and verdict is the right workflow.

6. Workflow — how to act on MLI / MLS

6.1 Daily — morning briefing

You don't need to read raw MLI/MLS. Read the fatigue type tag on each flagged player. The tag tells you what to modify:

- “mechanical fatigue” → swap the highest-decel drill for a technical / passing one
- “metabolic fatigue” → cut the long-distance running portion, keep the explosive work
- “global fatigue” → cut overall session intensity by ~20% or convert to recovery
- “normal” → train as planned

6.2 After a match

Both MLI and MLS will be high for most players the next morning — that's expected. The Composite Load Concern engine and the AI Summary now tag this as “post-match — expected” automatically (since we wired in match-day awareness). Don't panic if every player's fatigue type tag reads “global fatigue” the morning after a match.

6.3 During Return-to-Play

For a player coming back from injury, watch the trajectory of MLI specifically:

1. **Weeks 1–2** — MLI in the 30–50 range (LOW to MODERATE). Player is being protected — good.
2. **Weeks 3–4** — MLI rising into 50–65 range (MODERATE to HIGH). Body re-adapting to mechanical demand.
3. **Week 5+** — MLI hitting 70+ on individual sessions (VERY_HIGH). Player is back to full training tolerance.

A return-to-play that jumps from MLI 30 to MLI 75 in one session is a re-injury risk. Smooth progression is what you want.

6.4 Periodisation review

Look at team-average Residual MLI across the week. A team consistently in NORMAL band is in a healthy training rhythm. Multiple consecutive days where the team is in ELEVATED or CAUTION means you're carrying multi-day mechanical load — make sure the next 48 hours have recovery built in.

7. Common patterns

“Composite is high but MLI is low”

Either the metabolic arm or the RPE-ACWR arm is driving the composite. Read the fatigue type — if “metabolic fatigue”, the metabolic side is dominant. If “normal”, check the wellness check-in — the player may just be self-reporting tired without the load actually being high.

“MLI is HIGH but the player feels fine”

Two reads: 1. **Post-match (MD+1, MD+2)** — expected. Body is supposed to feel it. The player will feel it tomorrow even if not today. 2. **Mid-week** — the player is well-adapted to high mechanical load. Watch for delayed-onset soreness 24-48h later, but no immediate action.

“MLS is VERY_HIGH and MLI is LOW”

Pure metabolic day — typical of a positional/possession session with lots of high-speed running but limited braking or direction changes. Common for wingers and attacking midfielders. Action: prioritize hydration and sleep; mechanical strength work tomorrow is fine.

“MLI is LOW for everyone but Composite is high”

Sanity-check the ACWR (RPE arm) — if RPE is high but mechanical is low, it might be a technical session where players reported high effort but the load monitor didn’t catch it (e.g. a tactical walkthrough with lots of stop-start that doesn’t register as decels). Trust RPE in this case.

“MLI confidence is ‘low’ for several players”

Catapult sync had a partial day. Open /coach/load and check raw signal coverage. Usually self-corrects within 24 hours when the next sync runs.

8. Sport-science references

- **Vanrenterghem et al. 2017** (*Sports Medicine*) — Training load and mechanical injury risk in team sports
- **McBurnie, Harper, Jones & Dos’Santos 2022** (*Sports Medicine*) — Deceleration framework (drives the DSS weighting in MLI)
- **di Prampero et al. 2005** (*J Exp Biol*) — Sprint running energetics (basis for Catapult’s metabolic power)
- **Brown et al. 2016** (*J Sci Med Sport*) — GPS-derived metabolic power validation vs gas analysis (the Brown correction we apply)
- **Halson 2014** (*Sports Med*) — Training load monitoring framework

- **Bourdon et al. 2017** (*Int J Sports Physiol Perform*) — Monitoring athlete training loads consensus statement
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Last updated: 29 April 2026 · MicroPulse Performance Intelligence Platform